

From Cycle Space to Cycle Manifold: Limits and Achievability of Blind False Data Injection Attacks

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Abstract—A false data injection attack (FDIA) can change the estimated grid state while evading a residual-based bad-data detector (BDD). Existing blind attacks learn a low-rank measurement subspace, but this algebraic view does not state the physical grid constraints that make an attack stealthy or the minimum information needed to recover the complete attack space. Under the connected direct-current (DC) branch-flow model, we show that the residual-sensitive subspace of the noiseless orthogonal test is exactly the weighted cycle space. Its orthogonal complement is therefore the complete stealthy attack space, making weighted cycle-space knowledge both necessary and sufficient for complete blind FDIA. This space identifies the topology only up to 2-isomorphism and the relative cycle-edge parameters only up to one scale per biconnected component; bridge parameters are neither identified nor required. We then formulate a computationally unconstrained benchmark and a tractable measurement-only reconstruction method. Experiments on IEEE systems compare BDD bypass rate at a 95% nominal-acceptance threshold against state impact. As a compact alternating-current (AC) extension, we characterize feasible branch P/Q measurements by a cycle manifold and demonstrate topology-assisted manifold fitting and measurement generation on a graphics processing unit (GPU). In the lossless fixed-voltage small-angle limit, the normal space of the active-power slice reduces to the DC weighted cycle space.

Index Terms—Blind false data injection attack, cycle space, direct-current power flow, power system state estimation, topology identification.

I. INTRODUCTION

Power system state estimation turns measurements into the bus states used for grid monitoring and control, while a residual-based bad data detector (BDD) rejects measurements that are inconsistent with the estimated state [1]. On the attacker's side, given sufficient system information about the power grid, a false data injection attack (FDIA) can bias the state without causing a corresponding increase in residuals, so the control center may accept an incorrect view of the grid [2], [3]. Luckily, the power grid's system information is well protected. So the attacker cannot easily inject a false measurement without an BDD alarm. The most feasible information for the attacker is the time series measurement. The measurement contains the normal operation of the power grid. Extracting useful information to craft a stealthy injection from normal measurement is called blind FDIA. blind FDIA

is the most realistic threat to the modern power grid because it requires minimal information.

Blind FDIA replaces knowledge of the system with historical measurements. Existing methods estimate the attack space through subspace learning [4], PCA [5], random-matrix perturbation [6], or matrix reconstruction [7]. These methods show that a low-rank measurement structure can support blind FDIA, but the learned object typically captures only a small subset of the attack directions rather than the complete attack space.

Besides these subspace methods, some topology and physics-related work is also proposed. Martin et al. [8] try to estimate the full system information using the method in [9]. The information required to estimate the full system is not only the branch power flow but also the state node voltage. Chin et al. [10] propose a linear single attack direction estimation method with the assumption that the difference of state deviation is small. Some deep learning and machine learning methods have also been proposed for blind FDIA. But these methods have a minor influence.

Since the early development of FDIA [1], a large body of research has proposed increasingly sophisticated approaches for constructing FDIA. Yet several fundamental questions remain insufficiently understood: What is the physical origin of FDIA stealthiness? What underlying structure gives residual-based BDD its protection capability? And what fundamentally determines the limits and achievability of FDIA? In this paper, we address these questions from a unified structural perspective and develop a deeper understanding of both BDD and FDIA. We show that the key lies in the network's cycle structure: in the DC model, this structure appears as the linear cycle space, while in the AC model, it generalizes to a nonlinear cycle manifold. From this viewpoint, residual protection, stealthy perturbations, and the fundamental limits of blind attacks can all be understood through the same underlying principle.

Cycle physics is established separately in the power system and graph literature. Cycle variables express power-flow consistency [11], [12], while the graph cycle space determines topology only up to 2-isomorphism [13], [14]. These works do not establish the relationship between FDIA and cycle structures.

In this paper, we establish the explicit physical constraints for both DC and AC power models. We reveal that the cycle structure is the fundamental component for both AC and AC models. In the DC model, the FDIA is constrained by the weighted cycle space. In the AC model, the FDIA is constrained by the cycle manifold. We prove that the weighted cycle space and cycle manifold are sufficient and necessary conditions to perform a complete FDIA for DC and AC models, respectively. We therefore study the limits and achievability of blind FDIA using these physical constraints. With these theories and studies, we not only provide a new tool for FDIA research but also offer new insights into power grid planning and protection.

The main contributions are as follows:

- **Physical origin of FDIA stealth.** We prove that the weighted cycle space is the exact¹ orthogonal complement of the complete attack space in the DC model. For the AC model, we propose the theory of cycle manifold and prove that it is the exact nonlinear constraint for the complete attack space. With linearization and a small voltage angle assumption, the cycle manifold can be transformed to the constraint in weighted cycle space.
- **Limits and achievability** Estimating the weighted cycle space and cycle manifold both need a cycle basis and a parameter estimation process. We develop a tractable measurement-only DC weighted cycle space reconstruction method. With topology and measurement information, we can obtain an upper bound for cycle space estimation. For the AC model, we show that it is almost impossible to estimate a correct cycle basis because it is a nonlinear combinatorial problem. With topology and measurement information, we develop a zero-bias cycle manifold estimation method. Both methods outperform the existing blind FDIA methods.
- **Planning and protection implications.** Given the limits and achievability, we provide several guidelines for power grid planning and protection.

Our earlier work uses cycle-space reconstruction for detection and studies its reconstruction and generalization errors [15]. Here, we use the cycle view to establish the physical origin of the FDIA, the minimum attacker information, the computational limit, and DC/AC attack realizations.

II. PRELIMINARIES

A. State estimation, BDD and blind FDIA

We represent the operational transmission grid by a graph $\mathbf{G} = \mathbb{G}(E, V)$ with $m = |E|$ branches and $n = |V|$ buses. $q = m - n + 1$ is the number of cycles in the grid.

Under the AC model, the reference-bus voltage magnitude and angle are fixed. Let $\vec{x} \in \mathbb{R}^{2n}$ contain the voltage magnitudes and angles. The measurements satisfy

$$\vec{z} = h(\vec{x}) + e, \quad (1)$$

where $h(\cdot)$ is the nonlinear AC measurement function and e is measurement noise. The weighted least-squares state estimate is

$$\hat{\mathbf{x}} = \arg \min_{\vec{x}} [\vec{z} - h(\vec{x})]^T \mathbf{R}^{-1} [\vec{z} - h(\vec{x})], \quad (2)$$

where $\mathbf{R}$ is the measurement-noise covariance. The BDD uses

$$J_{AC}(\vec{z}) = [\vec{z} - h(\hat{\mathbf{x}})]^T \mathbf{R}^{-1} [\vec{z} - h(\hat{\mathbf{x}})]. \quad (3)$$

It rejects the measurements when $J_{AC}(\vec{z}) > \tau$ and accepts them otherwise.

Under the DC approximation, the active branch-flow measurement on line (i, j) is

$$\mathbf{P}_{ij} = \frac{\theta_{ij}}{x_{ij}}, \quad (4)$$

where $\theta_{ij} = \theta_i - \theta_j$. Stacking all branch-flow measurements gives

$$\vec{z} = \mathbf{H}\theta + e \quad (5)$$

where $\theta \in \mathbb{R}^n$ is the voltage-angle state and $e \in \mathbb{R}^m$ is measurement noise. The Jacobian decomposes as

$$\mathbf{H} = \mathbf{A} \odot \mathbf{p} \quad (6)$$

where $\mathbf{p} \in \mathbb{R}^m$ contains line susceptances, $\odot$ denotes row-wise element-wise multiplication, and $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the topology incidence matrix:

$$\mathbf{A}_{ij} = \begin{cases} 1 & \text{if } v_j \text{ is the initial node of link } e_i, \\ -1 & \text{if } v_j \text{ is the terminal node of link } e_i, \\ 0 & \text{otherwise.} \end{cases} \quad (7)$$

The weighted least-squares state estimate is tested by the standard residual sensitivity matrix

$$\mathbf{S} = \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{R}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{R}^{-1}, \quad (8)$$

where $\mathbf{R}$ is the measurement-noise covariance. In the noiseless case, $\mathbf{S}_o = \mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T$ is the orthogonal projector onto $\mathbb{N}(\mathbf{H}^T)$. Hence an injected vector $\vec{z}_a$ evades the residual test whenever $\vec{z}_a \in \mathbb{R}_\rho(\mathbf{H}^T)$.

$\mathbf{H}$ has rank $n - 1$ for a connected grid after the reference angle is fixed, so $\dim \mathbb{N}(\mathbf{H}^T) = q$.

a) *Blind FDIA:* A blind FDIA attacker observes historical branch-flow measurements but does not know the system information. The attacker is successful if it can construct a nonzero vector $\vec{z}_a$ such that the attacked measurement $\vec{z} + \vec{z}_a$ remains below the BDD threshold while changing the estimated state. In the noiseless DC model, this means constructing $\vec{z}_a \in \mathbb{R}_\rho(\mathbf{H}^T)$ without direct access to $\mathbf{H}$. Existing blind methods estimate this row space statistically, for example, through subspace estimation, principal component analysis (PCA)-type factorization, or partial topology-assisted learning [4]–[8].

B. Cycle space and 2-isomorphism

$\mathbf{H}$ can be regarded as a weighted incidence matrix of $\mathbf{A}$. The cycle space property of $\mathbf{A}$ has been well studied in graph theory [16]–[18].

¹Exact means necessary and sufficient

Definition 1 (Cycle terminology). A cycle is a closed walk with no repeated vertices except that its first and last vertices are the same, represented by its edge set $c \subseteq E$. A spanning tree S_G is a cycle-free connected subgraph containing all buses. Edges outside S_G are chords C_G . Each chord together with S_G induces a fundamental cycle.

Definition 2 (Biconnected component). A biconnected component (also known as a 2-connected component or block) of an undirected graph G is a maximal subgraph in which the removal of any single node does not disconnect the subgraph. Equivalently, a biconnected component is a subgraph in which every pair of nodes is connected by at least two internally node-disjoint paths. In this paper, we denote a biconnected component by $\mathbb{B} = (V_{\mathbb{B}}, E_{\mathbb{B}})$, where $V_{\mathbb{B}}$ and $E_{\mathbb{B}}$ are the node and edge sets of the component, respectively.

Definition 3 (Cycle space). The cycle space of G is $C(G) = \mathbb{N}(\mathbf{A}^T) \subseteq \mathbb{R}^{m \times q}$. It is spanned by signed cycle-indicator vectors $\chi_c \in \mathbb{R}^m$, whose entries are +1 or -1 on oriented cycle edges and 0 otherwise. All possible cycle space of G forms the cycle space set $\mathcal{C}$.

There are q linearly independent cycles (that are fundamental cycles). We denote by F_C a fundamental cycle set associated with a spanning tree.

Definition 4 (2-isomorphic graphs). Two graphs are strictly 2-isomorphic if there is a one-to-one edge correspondence under which cycles correspond to cycles [13]. Equivalently, two graphs are 2-isomorphic iff they have the same cycle space [14].

III. MINIMUM PHYSICAL INFORMATION FOR COMPLETE FDIA

A. DC weighted cycle space

For a fixed Jacobian $\mathbf{H}$, define the noiseless stealthy attack space as

$$\mathcal{A}(\mathbf{H}) = \mathbb{R}_p(\mathbf{H}^T). \quad (9)$$

An attacker has complete FDIA capability if it can recover the entire subspace.

Definition 5 (Weighted cycle space). Let $\mathbf{D}_p = \text{diag}(p)$. For an oriented cycle c , its weighted signed indicator is

$$\chi_{c,w} = \mathbf{D}_p^{-1} \chi_c. \quad (10)$$

The weighted cycle space is

$$C_w(\mathbf{G}) = \text{span}\{\chi_{c,w} : c \text{ is a cycle of } \mathbf{G}\}. \quad (11)$$

Theorem 1 (DC weighted cycle space completeness). Assume that G is connected, every entry of p is nonzero, and $\mathbf{H} = \mathbf{D}_p \mathbf{A}$. The following statements are equivalent:

- 1) the complete stealthy attack space $\mathcal{A}(\mathbf{H})$ is known;
- 2) the weighted cycle space $C_w(\mathbf{G})$ is known;
- 3) a cycle basis and block-consistent relative line parameters on its cycle edges are known.

In particular,

$$\mathbb{N}(\mathbf{H}^T) = C_w(\mathbf{G}), \quad \mathcal{A}(\mathbf{H}) = C_w(\mathbf{G})^\perp. \quad (12)$$

For a cycle basis $F_C = \{c_1, \dots, c_q\}$ and any invertible $\mathbf{R} \in \mathbb{R}^{q \times q}$, let

$$\begin{aligned} \mathbf{N}_w &= [\chi_{c_1,w}, \dots, \chi_{c_q,w}], \\ \text{col}(\mathbf{N}_w \mathbf{R}) &= \text{col}(\mathbf{N}_w) = C_w(\mathbf{G}). \end{aligned} \quad (13)$$

The attacker requires $\text{col}(\mathbf{N}_w)$. $\mathbf{N}_w$ and $\mathbf{N}_w \mathbf{R}$ contain identical information.

Minimum refers to the necessary and sufficient conditions for recovering the entire subspace.

Proof. For every cycle c , $\mathbf{A}^T \chi_c = \mathbf{0}$. Hence

$$\mathbf{H}^T \chi_{c,w} = \mathbf{A}^T \mathbf{D}_p \mathbf{D}_p^{-1} \chi_c = \mathbf{0}. \quad (14)$$

The q fundamental-cycle indicators are independent. Multiplication by the invertible matrix $\mathbf{D}_p^{-1}$ preserves independence, while $\dim \mathbb{N}(\mathbf{H}^T) = q$. Therefore they form a basis of $\mathbb{N}(\mathbf{H}^T)$, which proves the first equality in eq. (12). The second follows by taking the orthogonal complement. Consequently, knowing $C_w(\mathbf{G})$ gives the entire attack space, and knowing the entire attack space gives $C_w(\mathbf{G})$. This proves the equivalence of the first two statements.

On each cycle, the nonzero entries of its weighted indicator determine the relative line parameters. Shared branches align these ratios within each biconnected component. Thus, the weighted cycle space gives the third statement. Conversely, the cycle basis and block-consistent relative parameters form the weighted indicators in eq. (10). Their span is $C_w(\mathbf{G})$. Therefore, the third statement also gives the second. Finally, right multiplication by an invertible matrix changes only the basis coordinates, proving eq. (13). $\square$

Corollary 2 (Identifiability). Complete FDIA does not require distinguishing graphs within a 2-isomorphism class, identifying the parameter scale of a biconnected component, or identifying bridge parameters.

Proof. $C_w(\mathbf{G})$ determines the unweighted cycle space and therefore the topology only up to 2-isomorphism. Within each biconnected component $\mathbb{B}$, the weighted cycle vectors determine relative line parameters, but not their scale:

$$p_{\mathbb{B}} = k_{\mathbb{B}} \hat{p}_{\mathbb{B}}. \quad (15)$$

This scale does not change $C_w(\mathbf{G})$. A bridge is in no cycle, so its parameter does not appear in the weighted cycle space and is not needed for complete FDIA. $\square$

The cycle basis is also not unique. Every valid basis spans the same weighted cycle space in the noiseless case, as shown by eq. (13).

B. AC cycle manifold

Let $s_e = P_e + jQ_e$ be the complex power stored at the from end of branch $e = (i, j)$, and let $u_i = |V_i|^2$. With the branch current $I_e = Y_{ff,e} V_i + Y_{ft,e} V_j$, the conjugate branch-power equation gives

$$\overline{s_e} = Y_{ff,e} u_i + Y_{ft,e} \overline{V_i} V_j. \quad (16)$$

where $Y_{ff,e}$ is the branch's from-end self-admittance. $Y_{ft,e}$ is the transformer or mutual admittance from the to-end voltage

V_j to the from-end current I_e . Hence, when $V_i \neq 0$ and $Y_{ft,e} \neq 0$, the branch voltage ratio is

$$\begin{aligned} \rho_e &\equiv \frac{V_j}{V_i} = A_e + B_e \frac{\bar{s}_e}{u_i}, \\ A_e &= -\frac{Y_{ff,e}}{Y_{ft,e}}, \quad B_e = \frac{1}{Y_{ft,e}}. \end{aligned} \quad (17)$$

Let $\mathbf{C} \in \{-1, 0, 1\}^{q \times m}$ be an oriented fundamental-cycle matrix. Voltage ratios telescope around each cycle, so every feasible AC measurement satisfies

$$\mathcal{H}_k \equiv \prod_{e=1}^m \rho_e^{C_{ke}} - 1 = 0, \quad k = 1, \dots, q. \quad (18)$$

Each equation in eq. (18) is complex and therefore gives two real constraints.

Theorem 3 (AC cycle manifold completeness). *Fix the reference-bus voltage magnitude and angle. Suppose that the grid is connected, the bus voltages and $Y_{ft,e}$ are nonzero, and the full AC measurement Jacobian has rank $2n - 2$, and the P/Q Jacobian on every spanning tree is nonsingular. The noiseless measurement set is a manifold of dimension $2n - 2$. The chord measurements obey a map:*

$$\mathbf{z}_C = F_T(\mathbf{z}_T), \quad \mathbf{z}_C \in \mathbb{R}^{2q}. \quad (19)$$

Zero AC residual, eq. (19), and the q complex cycle equations in eq. (18) are equivalent.

Proof. Let h_T and h_C denote the tree and chord parts map between the AC measurement and state. The Jacobian of $h_T : \mathbb{R}^{2n-2} \rightarrow \mathbb{R}^{2n-2}$ is nonsingular. The tree measurements are sufficient to determine the voltage state $\vec{x} = g_T(\mathbf{z}_T)$. and

$$F_T(\mathbf{z}_T) = h_C(g_T(\mathbf{z}_T))$$

is a chord completion map.

Necessity follows because $\rho_e = V_j/V_i$, the voltage ratios multiply by one around every closed cycle.

For sufficiency, consider measurements $(\mathbf{z}_T, \mathbf{z}_C)$ that satisfy all fundamental-cycle equations. First recover $\vec{x} = g_T(\mathbf{z}_T)$. The tree measurements then give the tree-edge voltage ratios of this state. Each fundamental cycle contains one chord $c = (i, j)$ and the unique tree path between buses i and j . Equation (17) gives

$$\bar{s}_c = u_i \frac{\rho_c - A_c}{B_c}.$$

Because $u_i > 0$ and $B_c = 1/Y_{ft,c} \neq 0$, this ratio determines one unique chord power s_c . Every chord measurement equals $h_C(g_T(\mathbf{z}_T))$, so $\mathbf{z}_C = F_T(\mathbf{z}_T)$ and the complete measurement vector is produced by one AC state. This proves the equivalence with zero AC residual. $\square$

Corollary 4 (Identifiability). *Under the assumptions of theorem 3, The branch parameters must be consistent across cycles in the same biconnected component. Their voltage-scaling ambiguity and alignment across different components cannot be identified. Bridge parameters are also not separately identifiable or required.*

Proof. Every cycle is contained in one biconnected component, while a bridge is in no cycle. The rows of $\mathbf{C}$ can be grouped by biconnected component, and $C_{ke} = 0$ for every bridge e . Equation (18) consequently requires branch parameters only among cycles in the same component and places no cycle constraint on a bridge parameter.

The parameters are subject to a voltage-scaling ambiguity. For any nonzero bus scalars h_i , with $h_{\text{ref}} = 1$, set

$$W_i = h_i V_i, \quad A'_e = \frac{h_j}{h_i} A_e, \quad B'_e = h_j \bar{h}_i B_e, \quad e = (i, j). \quad (20)$$

Since $|W_i|^2 = |h_i|^2 u_i$,

$$A'_e + B'_e \frac{\bar{s}_e}{|W_i|^2} = \frac{h_j}{h_i} \left(A_e + B_e \frac{\bar{s}_e}{u_i} \right) = \frac{W_j}{W_i}. \quad (21)$$

The factors h_j/h_i cancel around every cycle. This transformation changes the coefficient representation, including parameters on bridges and between components, without changing the cycle manifold. By theorem 3, an attacked measurement $\mathbf{z} + \mathbf{z}_a$ has zero AC residual if and only if, for some sufficiently small $\Delta \mathbf{z}_T$,

$$\mathbf{z}_a = \left[F_T(\mathbf{z}_T + \Delta \mathbf{z}_T) - F_T(\mathbf{z}_T) \right]. \quad (22)$$

For $\Delta \mathbf{z}_T \neq \mathbf{0}$, the parameters inverse g_T gives a different voltage state. The blockwise cycle manifold is the identifiable object required for complete FDIA. $\square$

The DC result in theorem 2 is the linear example: 2-isomorphic graphs, biconnected-component parameter scales, and bridge parameters are indistinguishable because they leave the weighted cycle space, and therefore the complete attack space, unchanged.

C. Connection between AC cycle manifold and DC cycle space

Under the flat voltage, small angle assumption, $\rho_e \approx 1 - jx_e P_e$. Taking the first-order logarithm of eq. (18) yields

$$\mathbf{C} \mathbf{D}_x \delta \mathbf{P} = \mathbf{0}, \quad \mathbf{D}_x = \text{diag}(\mathbf{x}) = \mathbf{D}_p^{-1}. \quad (23)$$

Let $\mathcal{M}_{AC}^P$ denote this fixed-voltage active power of the AC cycle manifold, with the flat point represented by $\mathbf{P} = \mathbf{0}$. The columns of $\mathbf{D}_x \mathbf{C}^T$ are the weighted cycle indicators in eq. (10). Its tangent and normal spaces are

$$\begin{aligned} T_0 \mathcal{M}_{AC}^P &= \mathbb{N}(\mathbf{C} \mathbf{D}_x) = C_w(\mathbf{G})^\perp = \mathcal{A}(\mathbf{H}), \\ N_0 \mathcal{M}_{AC}^P &= \text{col}(\mathbf{D}_x \mathbf{C}^T) = C_w(\mathbf{G}). \end{aligned} \quad (24)$$

The DC cycle space is the active-power tangent space of the AC cycle manifold. The DC cycle-space model is the first-order approximation of the AC cycle manifold.

A blind FDIA attacker only has information about the measurement. Learn cycle space (CS) and cycle manifold (CM) from measurement is a physically informed machine learning approach. In the next two sections, we are going to propose the blind FDIA realization method based on CS and CM. There are two stages: cycle-based estimation and

parameter estimation. Cycle-based estimation is a combinatorial problem. Solving such a problem brutally is NP-hard and requires almost infinite time to optimize on current electronic computers. We are going to realize two blind FDIA: Computationally Constrained Blind FDIA Realization, which requires only the measurement information. Computationally Unconstrained Blind FDIA Realization, which requires the measurement and topology information.

IV. COMPUTATIONALLY CONSTRAINED BLIND FDIA REALIZATION

In this section, the given information is measurement-only, which is common in classic blind FDIA.

A. DC realization

Let

$$\mathbf{Z} = [\bar{z}_1, \dots, \bar{z}_T] = \mathbf{H}\mathbf{X} + \mathbf{E} \in \mathbb{R}^{m \times T} \quad (25)$$

contain all branch-flow measurement channels. We assume $T > m$. Let $r = n - 1$. The output is a full-column-rank weighted cycle matrix

$$\hat{\mathbf{N}}_C = [\hat{n}_1, \dots, \hat{n}_q] \in \mathbb{R}^{m \times q}, \quad \hat{\mathbf{N}}_C^T \mathbf{Z} \approx \mathbf{0}. \quad (26)$$

Its column space estimates $\mathbb{N}(\mathbf{H}^T)$, while $\mathbb{N}(\hat{\mathbf{N}}_C^T)$ estimates the physical measurement and attack space $\mathbb{R}_\rho(\mathbf{H}^T)$. The implementation uses root mean square (RMS) normalization, a rank- r singular value decomposition (SVD), pivoted QR, sparse support screening, and total least squares (TLS) fitting.

1) *RMS normalization and algebraic-tree recovery*: For each branch e , we define its uncentered RMS scale by

$$s_e = \left(\frac{1}{T} \|\mathbf{Z}_{e,:}\|_2^2 \right)^{1/2}, \quad (27)$$

$$\mathbf{D}_s = \text{diag}(s), \quad \tilde{\mathbf{Z}} = \mathbf{D}_s^{-1} \mathbf{Z}.$$

where all s_e must be nonzero. Equation (27) is a branchwise scale correction.

We decompose $\tilde{\mathbf{Z}} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ such that $\mathbf{U}_r \in \mathbb{R}^{m \times r}$ contain the first r left singular vectors. Apply column-pivoted QR to the branch coordinates:

$$\mathbf{U}_r^T \mathbf{P}_\pi = \mathbf{Q}\mathbf{R}, \quad \hat{\mathbf{S}}_G = \{e_{\pi(1)}, \dots, e_{\pi(r)}\}. \quad (28)$$

The remaining set $\hat{Q} = E \setminus \hat{\mathbf{S}}_G$ contains q candidate chords. The implementation processes these chords in increasing branch-index order. In noiseless data, the selected r independent branches form a spanning tree. In noisy data, Column-pivoted QR selects the most independent branches.

2) *Cycle basis recovery*: For each $e \in \hat{Q}$, the normalized chord measurement is regressed on all normalized tree measurements:

$$\tilde{\mathbf{Z}}_{e,:}^T \approx \tilde{\mathbf{Z}}_{\hat{\mathbf{S}}_G,:}^T \beta_e. \quad (29)$$

We use a no-intercept Lasso path and select its point by the BIC. The nonzero entries of β_e define the screened tree set $K_e \subseteq \hat{\mathbf{S}}_G$. If fewer than two branches are selected, ordinary least squares is used, and the two largest parameters are retained. This guarantees a chord candidate with at least two tree branches.

On $S_e^{\text{scr}} = \{e\} \cup K_e$, we compute the unit TLS vector

$$\mathbf{w}_e^{\text{scr}} \in \arg \min_{\|\mathbf{w}\|_2=1} \|\mathbf{w}^T \tilde{\mathbf{Z}}_{S_e^{\text{scr},:}}\|_2^2. \quad (30)$$

The branches in K_e are ordered by decreasing corresponding magnitude in $\mathbf{w}_e^{\text{scr}}$. Equal magnitudes are ordered by branch index. Let this order be $k_{e,1}, \dots, k_{e,|K_e|}$ and define

$$C_{e,j} = \{e, k_{e,1}, \dots, k_{e,j}\}, \quad j = 2, \dots, |K_e|. \quad (31)$$

For each support, the TLS residual and explicit BIC score are

$$\rho_{e,j} = \sigma_{\min}^2(\tilde{\mathbf{Z}}_{C_{e,j},:}), \\ \text{BIC}_{e,j} = T \log \left[\max \left(\frac{\rho_{e,j}}{T}, \varepsilon_{\text{tiny}} \right) \right] + |C_{e,j}| \log T, \quad (32)$$

where $\varepsilon_{\text{tiny}}$ is the machine positive-normal threshold.

The preliminary cycle $S_e^{(0)}$ is the candidate with the smallest score. Thus, the BIC is used twice: first to screen the Lasso path and then to choose the TLS prefix length.

3) *Weighted-constraint fitting and support refinement*: For any selected cycle S , let

$$\mathbf{w}_e(S) \in \arg \min_{\|\mathbf{w}\|_2=1} \|\mathbf{w}^T \tilde{\mathbf{Z}}_{S,:}\|_2^2. \quad (33)$$

This is the smallest left singular vector of $\tilde{\mathbf{Z}}_{S,:}$. To return to the original branch-flow coordinates, embed it in $\mathbb{R}^m$ as

$$\bar{n}_e(S)[\ell] = \begin{cases} \mathbf{w}_e(S)[\ell]/s_\ell, & \ell \in S, \\ 0, & \ell \notin S, \end{cases} \quad (34) \\ \bar{n}_e(S) = \frac{\bar{n}_e(S)}{\|\bar{n}_e(S)\|_2}.$$

The sign is fixed by making the parameter on the smallest-index cycle branch non-negative. This convention does not change the cycle equation. Stacking $\bar{n}_e(S_e^{(0)})$ gives the preliminary matrix $\hat{\mathbf{N}}_C^{(0)}$.

We then apply the fixed raw-coordinate threshold $\tau = 10^{-2}$:

$$\hat{c}_e = \{\ell \in E : |\hat{n}_e^{(0)}(\ell)| > \tau\}. \quad (35)$$

Equation (33) is solved again on each $\hat{c}_e$, and eq. (34) gives the final $\bar{n}_e$. We define $\hat{F}_C = \{\hat{c}_e : e \in \hat{Q}\}$ such that the final matrix is accepted only if

$$\text{rank}(\hat{\mathbf{N}}_C) = q. \quad (36)$$

The threshold is applied after the raw-coordinate conversion and unit normalization. The chord is not forced to remain, and a refined cycle can contain only two branches. Hence eq. (36) guarantees independent constraints. Also,

$$\hat{E}_{\text{no-cycle}} = E \setminus \bigcup_{e \in \hat{Q}} \hat{c}_e \quad (37)$$

is the set of branches absent from every recovered cycle. It equals the graph-bridge set when the recovered cycle basis link set is correct.

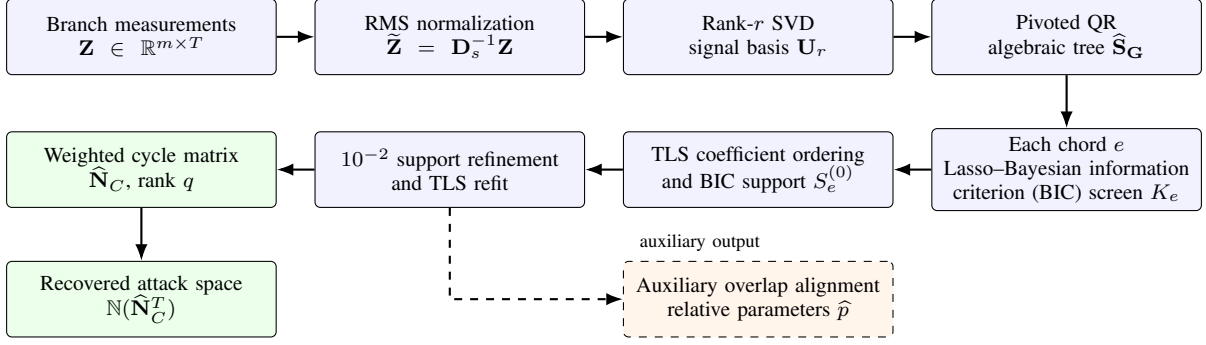


Figure 1. Measurement-only cycle-space recovery pipeline.

4) *Parameter estimation*: If $\hat{c}$ is a true cycle, then its exact weighted constraint obeys

$$\vec{n}_{\hat{c}} = \beta_{\hat{c}} \mathbf{D}_{\hat{c}}^{-1} \chi_{\hat{c}}, \quad \beta_{\hat{c}} \neq 0. \quad (38)$$

Thus, the cycle gives the branches, the signs give an orientation up to one global sign, and the magnitudes give relative line parameters:

$$\frac{p(\ell_1)}{p(\ell_2)} = \frac{|\vec{n}_{\hat{c}}(\ell_2)|}{|\vec{n}_{\hat{c}}(\ell_1)|}, \quad \ell_1, \ell_2 \in \hat{c}. \quad (39)$$

We also return a separate parameter estimate. It groups recovered cycles by shared branches and estimates a new smallest-eigenvector constraint $\vec{n}_c$ from $\mathbf{Z}_{c,:}; \mathbf{Z}_{c,:}^T$ for each cycle. If cycles c_i and c_j share $J_{ij} \neq \emptyset$, their coefficient scales are aligned by

$$\gamma_{ij} = \frac{1}{|J_{ij}|} \sum_{\ell \in J_{ij}} \frac{|\vec{n}_{c_i}(\ell)|}{|\vec{n}_{c_j}(\ell)|}, \quad \vec{n}_{c_j} \leftarrow \gamma_{ij} \vec{n}_{c_j}. \quad (40)$$

After propagating these ratios through each bi-connected component, the inverse coefficient magnitudes yield $\hat{p}$ up to a scale factor.

5) *Recovered attack space and residual*: We define:

$$\begin{aligned} \mathcal{W}_{\text{pr}} &= \text{col}(\hat{\mathbf{N}}_C), & \mathbf{P}_{\text{pr}} &= \hat{\mathbf{N}}_C \hat{\mathbf{N}}_C^+, \\ \hat{\mathbf{A}} &= \mathbf{N}(\hat{\mathbf{N}}_C^T) = \mathcal{W}_{\text{pr}}^\perp. \end{aligned} \quad (41)$$

For any $\mathbf{u} \notin \mathcal{W}_{\text{pr}}$ and $\alpha > 0$, a recovered attack direction is

$$\vec{z}_a = \alpha \frac{(\mathbf{I} - \mathbf{P}_{\text{pr}}) \mathbf{u}}{\|(\mathbf{I} - \mathbf{P}_{\text{pr}}) \mathbf{u}\|_2}, \quad (42)$$

such that $\hat{\mathbf{N}}_C^T \vec{z}_a = \mathbf{0}$.

For test data $\mathbf{Z}_{\text{test}}$, we report the raw cycle residual

$$\mathbf{E}_C = \hat{\mathbf{N}}_C^T \mathbf{Z}_{\text{test}}. \quad (43)$$

Since the unit-norm columns are not generally orthogonal, the norm of eq. (43) also depends on the basis of conditioning.

The realization uses no grid topology, incidence matrix, Jacobian, line parameters, or noise covariance. Its main operations are one rank- r SVD, one rank-revealing QR factorization (RRQR), q Lasso paths, and at most $q(r-1)$ small TLS evaluations. The main failure comprises three-coordinate errors, omitted Lasso branches, the fixed 10^{-2} threshold, removal of the chord during refinement, and independent final-cycle fits that need not share consistent line parameters.

Algorithm 1: Measurement-only cycle-space realization

Input: $\mathbf{Z}$, branch set E , number of buses n

Output: $\hat{\mathbf{S}}_G, \hat{F}_C, \hat{\mathbf{N}}_C$, auxiliary $\hat{p}$

- 1 Compute $\tilde{\mathbf{Z}}$ using eq. (27);
 - 2 Compute $\hat{\mathbf{S}}_G$ by rank- r SVD and pivoted QR using eq. (28);
 - 3 **for** $e \in \hat{Q}$ **do**
 - 4 Screen K_e by no-intercept Lasso–BIC using eq. (29);
 - 5 If $|K_e| < 2$, retain the two largest ordinary least-squares parameters;
 - 6 Order K_e by the TLS vector in eq. (30);
 - 7 Select $S_e^{(0)}$ by eqs. (31) and (32);
 - 8 Fit the preliminary raw-coordinate constraint by eqs. (33) and (34);
 - 9 **end**
 - 10 Refine every support by eq. (35) and refit its constraint;
 - 11 Verify eq. (36);
 - 12 Estimate auxiliary relative parameters using eq. (40);
 - 13 **return** $\hat{\mathbf{S}}_G, \hat{F}_C, \hat{\mathbf{N}}_C, \hat{p}$;
-

B. AC limitation

Because the AC CM is nonlinear, we can not find a shortcut to extract the cycle basis under AC.

V. COMPUTATIONALLY UNCONSTRAINED BLIND FDIA REALIZATION

In this section, the given information is measurement plus topology.

A. DC Realization

There are q fundamental cycles for a full rank null space estimation. The target to optimize is:

$$\min_{\hat{\mathbf{N}}_C} \|\hat{\mathbf{N}}_C^T \mathbf{Z}\|^2 \quad (44)$$

$$\text{s.t. } \|\hat{\mathbf{N}}_C\|^2 = 1, C \in \mathbb{C}. \quad (45)$$

This problem was solved in [15], where proves the minimum cycle basis achieve the optimal generalization error. The minimum cycle basis realization of DC unconstrained is the upper bound of DC blind FDIA.

B. AC realization

1) *Gauged Tree–Chord Cycle Manifold Fitting*: The AC realization assumes that the grid topology, the direction of each branch measurement, and the reference bus are known, while branch admittances and voltage states are not supplied. A spanning tree S is chosen to fix the fundamental cycles. Let l be the number of bridge branches in the graph. According to eq. (17), directly optimizing the AC manifold requires to estimate A_e, B_e and the voltage magnitude u_i . Directly estimating the voltage magnitude is unacceptable because it contributes $(n-1)T$ parameters. We proposed Gauged Tree–Chord Cycle Manifold Fitting (GTCM) method, which only estimate a variant of B_e . A_e is eliminated using gauge method. An equivalent variant of voltage magnitude is computed during the optimization, instead of working as optimization variables.

We introduce a gauged voltage at every bus:

$$V'_i = h_i V_i. \quad (46)$$

The branch power equation becomes:

$$\frac{V'_j}{V'_i} = \frac{h'_j}{h'_i} A_e + h'_j h'_i B_e \frac{\bar{s}_e}{|V'_i|^2}. \quad (47)$$

We define the variants of A_e, B_e as A'_e, B'_e :

$$A'_e = \frac{h_j}{h_i} A_e, B'_e = h_j h_i B_e. \quad (48)$$

By imposing the constraint $\frac{h_j}{h_i} A_e = 1$, $A'_e = 1$ holds everywhere for branches on a spanning tree S because there are $n-1$ free A_e parameters on a tree and there is a unique path for each branch to the reference branch.

Then we can compute the gauged voltage magnitude from the reference branch to every tree branch. For a tree branch $e = (i, j)$ traversed in the direction, V'_i is already known, so:

$$V'_j = 1 + B'_e \frac{\bar{s}_e}{V'_i} \quad (49)$$

If a branch orientation is opposite to the tree traversal, the implementation solves a quadratic equation for the unknown sending-end V_i and chooses the high-voltage root:

$$V'_j = \frac{V'_i + \sqrt{V'^2_i - 4B'_e \bar{s}_e}}{2} \quad (50)$$

We can compute all branch gauged voltage magnitude following the tree traversal.

Now take the chord. The tree ratios in this cycle are already known. According to the cycle manifold constraint in eq. (18), for a chord $c = (i, j) \in \mathcal{C}$ we define:

$$S_c = \prod_{e \in S} \rho_e^{C_{ke}}. \quad (51)$$

The chord ratio is therefore:

$$\rho_c = S_c^{-1} \quad (52)$$

The chord power flow is:

$$\bar{s}_c = \frac{|V_i|^2}{K_c} (\rho_c - A_c). \quad (53)$$

Define:

$$d_c = \frac{|V_i|^2}{K_c}, \quad g_c = d_c \rho_c. \quad (54)$$

Then:

$$\bar{s}_c = g_c - d_c A_c. \quad (55)$$

Suppose we have T training samples. For every chord c , we can get the closed-form solution of A_c :

$$\hat{A}_c = \frac{\sum_{r=1}^T d_c^{(r)} (g_c^{(r)} - \bar{s}_c^{(r)})}{\sum_{r=1}^T \|d_c^{(r)}\|^2} \quad (56)$$

Then we project the power back:

$$\hat{s}_c^{(r)} = g_c^{(r)} - d_c^{(r)} \hat{A}_c. \quad (57)$$

The optimization is established between the predict and measurement chord power:

$$\min_{B'} \frac{1}{2} \sum_{r=1}^T \sum_{c \in \mathcal{C}} \left\| \frac{\hat{s}_c^{(r)} - \bar{s}_c^{(r)}}{\sigma_c} \right\|^2 \quad (58)$$

2) *Attack generation*: In this section, we use the learned parameters to construct the attack. We first generate zero-mean Gaussian noise for the tree measurement $s_S^a = s_S + r \frac{\delta_S}{\|\delta_S\|^2}$. Where δ_S is generated under an independent normal distribution. r is a scaling factor controlling the attack magnitude. By filling in the learned parameters, we compute the chord measurement s_C^a by following eq. (46) to eq. (46). The fit and batched generation use double-precision CUDA computation on a graphics processing unit (GPU).

VI. DEFENDER-SIDE IMPLICATIONS

A. Grid planning

B. Design Principles for Defending Against Blind FDIA

The detection bound in the cycle space detector (CSD) framework [15] is optimal precisely because cycle-space recovery is the attacker's minimum requirement. In this sense, CSD is dual to the necessity theorem: it measures how far the attacker is from the only structure that can support uniformly stealthy blind FDIA.

A moving-target defense defeats the attacker if and only if the perturbation changes the recovered 2-isomorphism class or its weighted cycle-space realization. Perturbations that leave the 2-isomorphism class unchanged may change a nominal topology diagram, but they do not remove the attacker's sufficient information.

Measurement design should therefore restrict the attacker's ability to infer the weighted cycle space. Meter placement, aggregation, protected channels, and deliberate withholding of selected branch-flow streams are useful to the extent that they deny enough independent cycle information to reconstruct $\mathbb{N}(\mathbf{H}^T)$.

VII. EXPERIMENTS

A. Setup

We evaluate the proposed method on the IEEE 14-, 30-, 57-, and 118-bus test systems. For each grid, the first $T_{\text{train}} = m$ operating points form the training set. The clean branch RMS values s_e follow eq. (27), and the noisy training data are

$$\begin{aligned} \mathbf{Z}_{\text{train}}^{\text{noisy}} &= \mathbf{Z}_{\text{train}} + \mathbf{E}, \\ \mathbf{E}_{e,t} &\sim \mathcal{N}(0, \sigma_e^2), \quad \sigma_e = 0.1 \max(s_e, 10^{-8}). \end{aligned} \quad (59)$$

In the state-impact experiment, all methods are trained using a single noise realization generated with random seed 7. We compare seven DC attack-space estimates. *H known* uses $\text{col}(\mathbf{H})$. *CS unconstrained* uses the true minimum-cycle-basis supports but fits their weighted parameters from the same noisy data. *CS realization* uses the measurement-only estimator in Section IV-A. *PCA* estimates the standardized rank- r measurement space. *random matrix theory (RMT) perturbation* follows the random-matrix attack construction of [6], extended to the rank r used in this comparison. *Matrix reconstruction* implements the covariance-reconstruction attack of [7]. The *linear autoencoder* uses a rank- r linear encoder-decoder.

B. State impact versus 95%-calibrated pass rate

Let $\mathbf{D}_\sigma \equiv \text{diag}(\sigma)$, $\mathbf{H}_w = \mathbf{D}_\sigma^{-1} \mathbf{H}$, and let $\mathbf{Q}_\perp$ be an orthonormal basis of $\mathbb{N}(\mathbf{H}_w^T)$. From unattacked noisy trials, we set

$$\tau_{0.95} = Q_{0.95}\{J_i^{(0)}\}, \quad J_i = \|\mathbf{D}_\sigma \mathbf{Q}_\perp \mathbf{Q}_\perp^T \tilde{\mathbf{z}}_i\|_2^2, \quad (60)$$

where $\tilde{\mathbf{z}}_i$ is the whitened measurement. Thus 95% of nominal trials pass this fixed threshold.

For a method with attack basis $\mathbf{B} \in \mathbb{R}^{m \times r}$, reduced QR gives an orthonormal basis $\mathbf{Q}_B$ for $\text{col}(\mathbf{D}_\sigma^{-1} \mathbf{B})$. At strength γ , the paired trial direction is

$$\tilde{\mathbf{z}}_{a,i}(\gamma) = \gamma \mathbf{Q}_B \frac{\mathbf{Q}_B^T \mathbf{g}_i}{\|\mathbf{Q}_B^T \mathbf{g}_i\|_2}, \quad \mathbf{g}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (61)$$

The mean state impact and pass rate are

$$\begin{aligned} \bar{d}_{\text{DC}}(\gamma) &= \frac{1}{N\sqrt{r}} \sum_{i=1}^N \|\mathbf{H}_w^\dagger \tilde{\mathbf{z}}_{a,i}(\gamma)\|_2, \\ \pi(\gamma) &= \frac{1}{N} \sum_{i=1}^N \mathbf{1}\{J_i(\gamma) \leq \tau_{0.95}\}. \end{aligned} \quad (62)$$

Unlike a pass-rate-versus-threshold plot, this metric shows how much the state changes at the fixed detector operating point and avoids treating a negligible-impact attack as equally effective.

In Figure 2, CS unconstrained remains closest to the nominal 95% pass rate as state impact grows, showing the value of the correct cycle supports. CS realization is consistently the strongest measurement-only realization among the tested methods. The remaining subspace methods lose pass rate at smaller state impacts. The H-known marker is the nominal noise-only reference, rather than an attack-strength curve.

C. Null-space-overlap noise sensitivity

This experiment evaluates geometric recovery. We use IEEE-118, where

$$\begin{aligned} m &= 186, \quad n = 118, \quad r = 117, \quad q = 69, \\ T_{\text{train}} &= 186. \end{aligned} \quad (63)$$

The remaining 1,254 operating points are reserved for the attack experiment. For 10 logarithmically spaced levels $\alpha \in [10^{-2}, 10^{-1}]$, the noisy training matrix in trial j is

$$\mathbf{Z}_{\alpha,j} = \mathbf{Z}_{\text{train}} + \alpha \mathbf{D}_s \mathbf{\Xi}_j, \quad \mathbf{\Xi}_j[e, t] \sim \mathcal{N}(0, 1). \quad (64)$$

There are 20 trials at every level. Seed 17 is used, and the same $\mathbf{\Xi}_j$ is scaled across all levels and passed to all methods. Every method is re-estimated for every trial. A failed CS realization trial is assigned zero overlap, so its curve includes tree, support, coefficient, and rank failures.

Let $\mathbf{Q}_0 \in \mathbb{R}^{m \times q}$ be an orthonormal basis of $\mathbb{N}(\mathbf{H}^T)$ and let $\hat{\mathbf{Q}}$ span the estimated detector space. For the CS methods this is $\text{col}(\hat{\mathbf{N}}_C)$; for the signal-space methods it is the orthogonal complement of the estimated rank- r basis. We report

$$\eta = \frac{1}{q} \|\mathbf{Q}_0^T \hat{\mathbf{Q}}\|_F^2 = \frac{1}{q} \sum_{i=1}^q \cos^2 \theta_i, \quad (65)$$

where θ_i are the principal angles. Thus 100η measures average subspace alignment.

In Figure 3, CS unconstrained remains essentially at 100%, since only its parameters are refitted. CS realization remains above 99% through approximately 6% noise and falls to about 91% at 10% noise. At 10%, RMT perturbation, PCA, the linear autoencoder, and matrix reconstruction reach about 87.3%, 86.4%, 86.0%, and 84.5%, respectively. Hence the learned cycle structure achieves more robust subspace recovery over the evaluated range, although the final drop reflects discrete tree and support errors.

Figures 2 and 3 measure different properties. Figure 2 fixes one model learned at 10% noise and measures BDD acceptance against state impact in noise-normalized coordinates. Figure 3 re-estimates every method over 20 training-noise realizations and measures unwhitened Euclidean subspace geometry. Their method ordering therefore need not coincide.

D. AC cycle-manifold results

For AC, the threshold is the 95th percentile of the physical residual over held-out noisy measurements. If $\hat{V}_i, \hat{\theta}_i$ are the AC state estimates for a generated measurement and $\hat{V}_i^{(0)}, \hat{\theta}_i^{(0)}$ are those for its noisy base, the state impact is

$$\begin{aligned} d_i^{\text{AC}} &= \frac{1}{\sqrt{2(n-1)}} \left[\sum_{k \neq \text{ref}} (\Delta |V_{ik}|)^2 \right. \\ &\quad \left. + \sum_{k \neq \text{ref}} \text{wrap}(\Delta \theta_{ik} - \Delta \theta_{i,\text{ref}})^2 \right]^{1/2}. \end{aligned} \quad (66)$$

The angle difference removes the reference-angle gauge. The plotted point uses the mean of eq. (66) and the fraction whose

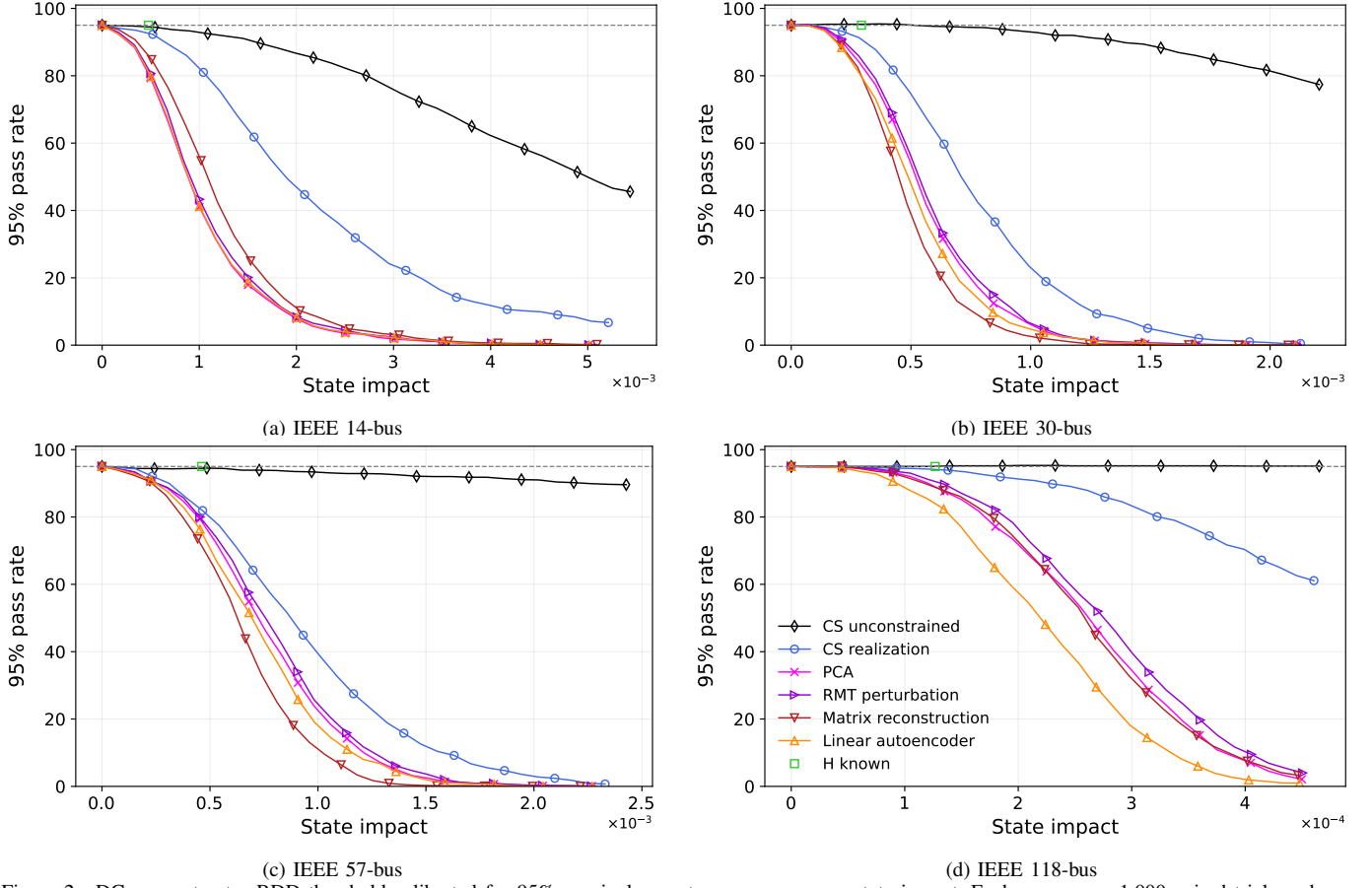


Figure 2. DC pass rate at a BDD threshold calibrated for 95% nominal acceptance versus mean state impact. Each curve uses 1,000 paired trials and one fixed model learned at 10% branch-relative training noise.

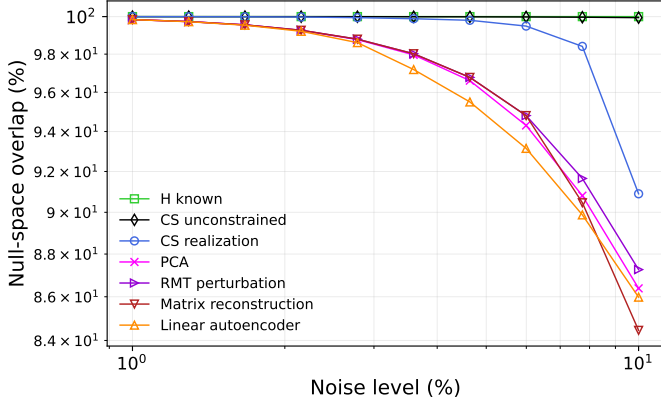


Figure 3. IEEE-118 null-space overlap versus branch-relative training noise. Curves show the mean over 20 paired trials at each level.

physical AC residual is below the fixed threshold, using the converged cohort across methods.

Figure 4 shows that the learned cycle manifold supports nontrivial state changes while retaining a high pass rate and is competitive with the tested subspace attacks on all four systems. Its gain is clearest on IEEE-30 and IEEE-57. Because direct completion overwrites the noisy base chords, CM may have a small nonzero impact even as the tree perturbation tends to zero. These results validate the topology-assisted

parameters construction; they do not establish measurement-only AC manifold recovery.

To separate systematic completion error from trialwise variance, let $e_{ic} = \hat{s}_{ic} - s_{ic}^{\text{clean}}$ on held-out chord c and define

$$b_{\text{chord}} = \text{baseMVA} \left(\frac{1}{q} \sum_{c=1}^q \left| \frac{1}{N} \sum_{i=1}^N e_{ic} \right|^2 \right)^{1/2}. \quad (67)$$

In Figure 5, CM unconstrained follows the system-known reference and its bias approaches numerical zero as measurement noise vanishes, whereas the compared data-driven estimators retain nonzero bias. Thus, under the correct topology, sufficient excitation, and a regular AC, CM unconstrained has the zero-bias noiseless limit. The nonlinear fit and the batch generation in these experiments are executed on a GPU.

VIII. CONCLUSION

This paper proves that weighted cycle-space recovery is the necessary and sufficient information requirement for blind FDIA from branch-flow measurements, yielding $\mathbf{H}$ only up to 2-isomorphism and one parameter scale per biconnected component. The AC extension shows that P/Q measurements are parametersly described by a necessary-and-sufficient cycle manifold and that this manifold can be fitted from topology and measurements to generate parametersly consistent data. Its lossless fixed-voltage small-angle linearization has the

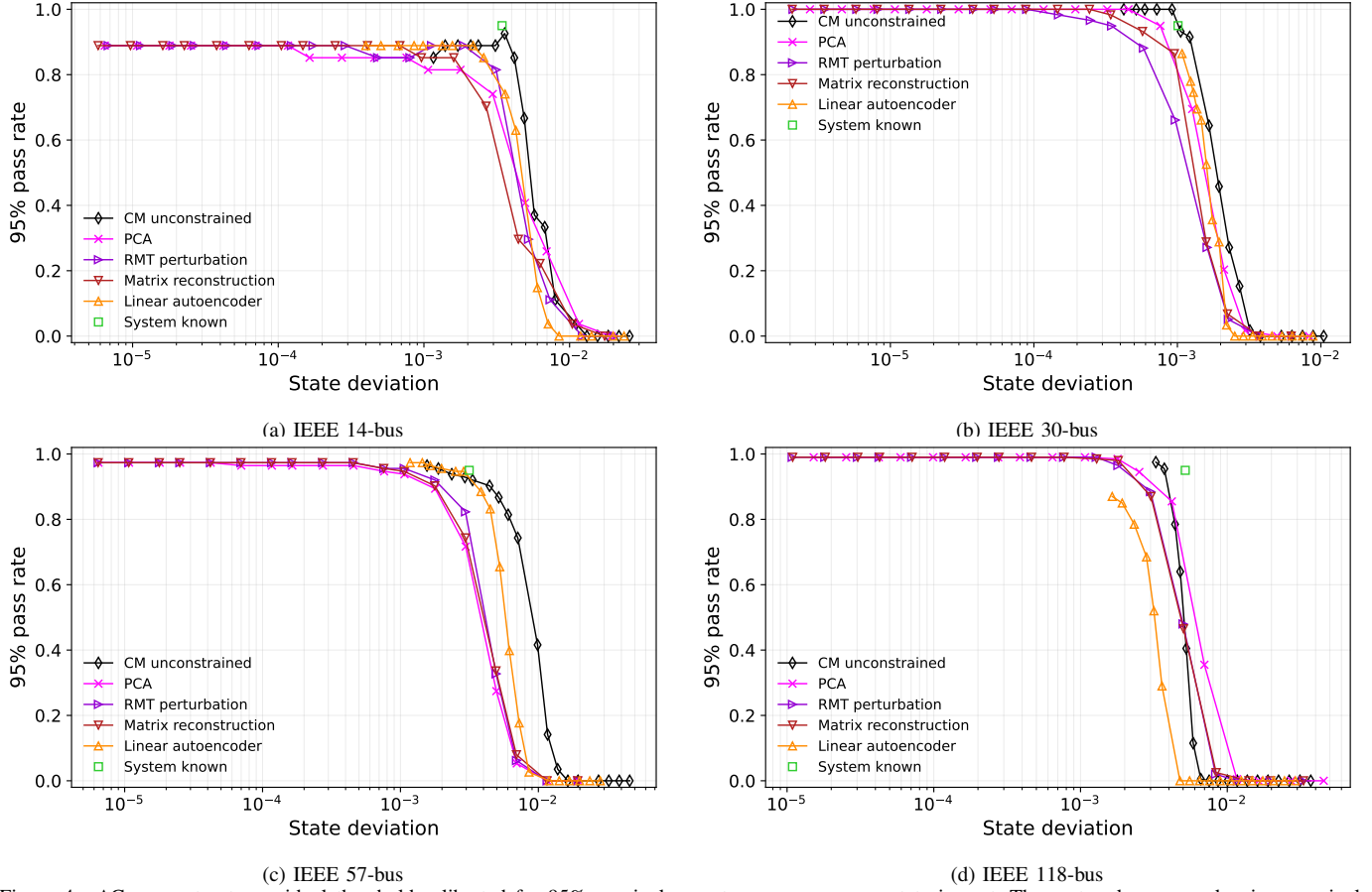


Figure 4. AC pass rate at a residual threshold calibrated for 95% nominal acceptance versus mean state impact. The system-known marker is a nominal noise-to-clean reference.

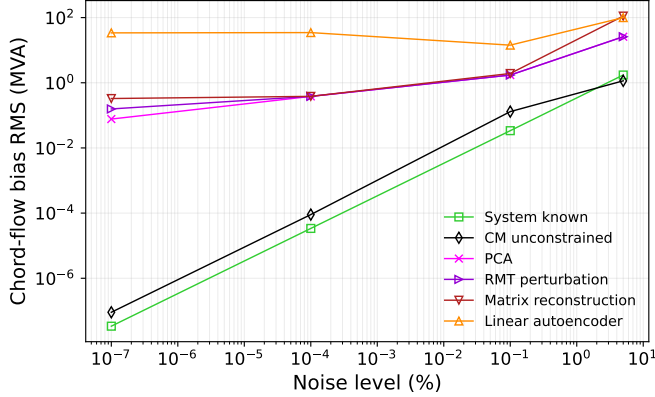


Figure 5. IEEE-118 chord-flow bias against measurement-noise level. CM unconstrained uses known topology but no physical admittances or voltage states.

DC stealthy space as its active-power tangent space and the weighted cycle space as its normal space. Learning the nonlinear fundamental-cycle structure without topology, and therefore deriving a measurement-only AC blind FDIA upper bound, remains open.

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